

## Experiment No. 1

### Test Case 1

Investigating the significance of Characteristic Impedance ( $Z_0$ ), and Propagation Constant ( $\gamma$ ) of a Transmission Line using the following parameters:

- a) Resistance =  $10 \Omega/\text{Km}$    b) Capacitance =  $8 \mu\text{F}/\text{Km}$    c) Inductance =  $3.5 \text{ mH}/\text{Km}$   
d) Conductance =  $4 \mu\text{mho}/\text{Km}$    e) Frequency =  $2 \text{ KHz}$

#### PROGRAM:

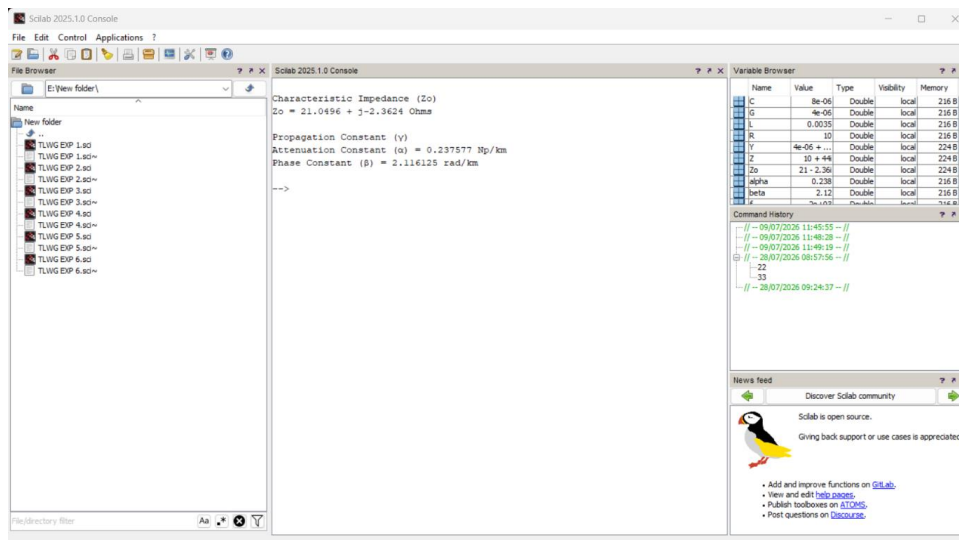
```
clear; clc;
R=10;L=0.0035;f=2*10^3;G=4*(10^-6);C=8*(10^-6);
w=2*pi*f; Z=R+(%i*w*L); Y=G+(%i*w*C); Zo=sqrt(Z/Y);
C1=round(real(Zo)); D=round(imag(Zo));
printf("-Zo = %f + j(%f)ohms\n",C1,D); P=sqrt(Z*Y);
a=real(P); a1=round(a*10000)/10000;
printf("-Attenuation constant a = %f neper/km\n",a1); b=imag(P);
b1=round(b*10000)/10000;
printf("-Phase constant b = %f radians/km",b1);
```

#### OUTPUT:

Constants:  $L = 3.5 \text{ mH}$ ;  $f = 2000 \text{ Hz}$ ;  $G = 4 \times 10^{-6} \text{ mho/km}$ ;  $C = 8 \times 10^{-6} \text{ F/km}$

| $Z_0$ (ohms) — Simulation | $\alpha$ (Np/km) | $\beta$ (rad/km) | $Z_0$ (ohms) — Theoretical | $\alpha$ (Np/km) | $\beta$ (rad/km) |
|---------------------------|------------------|------------------|----------------------------|------------------|------------------|
| $21.05 - j2.36$           | 0.23758          | 2.11613          | $21.05 - j2.36$            | 0.23758          | 2.11613          |

#### OUTPUT SCREENSHOT:



### Test Case 2

Examine the impact of resistance and inductance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line at a frequency  $f = 2.5$  KHz. Vary the resistance value from  $10\ \Omega$  to  $50\ \Omega$  at intervals of  $10\ \Omega$  with  $L$ ,  $G$  and  $C$  constant, then vary the inductance from  $3.5$  mH to  $11.5$  mH at intervals of  $2.5$  mH with  $R$ ,  $G$  and  $C$  constant.

#### PROGRAM:

```
clear; clc;
R=10;L=0.003;f=2.5*10^3;G=0.4*(10^-6);C=8*(10^-6);
w=2*pi*f; Z=R+(%i*w*L); Y=G+(%i*w*C); Zo=sqrt(Z/Y);
C1=round(real(Zo)); D=round(imag(Zo));
printf("-Zo = %f + j(%f)ohms\n",C1,D); P=sqrt(Z*Y);
a=real(P); a1=round(a*10000)/10000;
printf("-Attenuation constant a = %f neper/km\n",a1); b=imag(P);
b1=round(b*10000)/10000;
printf("-Phase constant b = %f radians/km",b1);
```

#### OUTPUT:

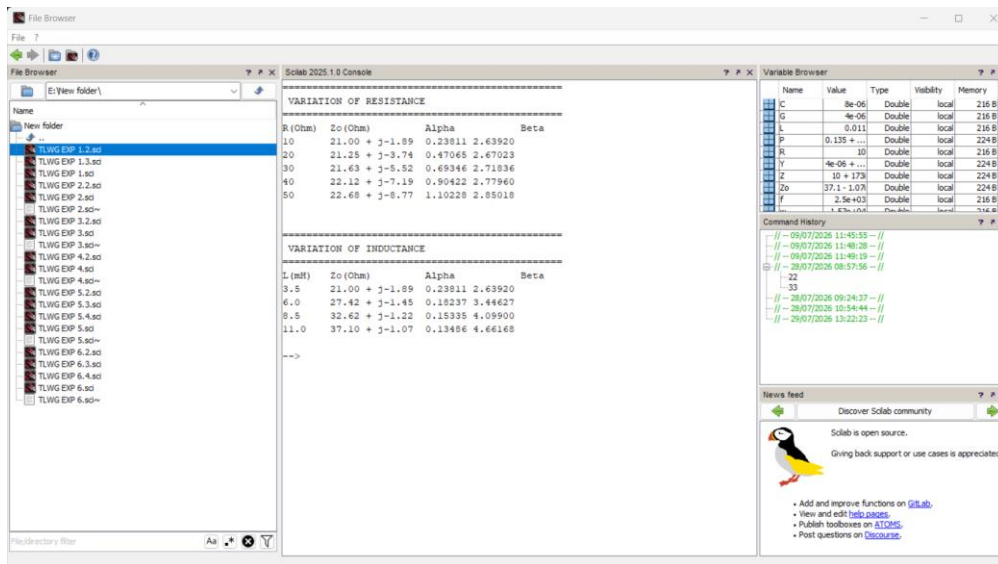
Variable R — Constants:  $L = 3$  mH;  $f = 2500$  Hz;  $G = 0.4 \times 10^{-6}$  mho/km;  $C = 8 \times 10^{-6}$  F/km

| R ( $\Omega$ ) | $Z_0$ (ohms)   | $\alpha$ (Np/km) | $\beta$ (rad/km) |
|----------------|----------------|------------------|------------------|
| 10             | 19.53 – j2.55  | 0.25600          | 1.96353          |
| 20             | 19.99 – j4.98  | 0.50016          | 2.00999          |
| 30             | 20.67 – j7.22  | 0.72580          | 2.07767          |
| 40             | 21.47 – j9.27  | 0.93162          | 2.15820          |
| 50             | 22.34 – j11.13 | 1.11922          | 2.24556          |

Variable L — Constants:  $R = 10\ \Omega$ ;  $f = 2500$  Hz;  $G = 0.4 \times 10^{-6}$  mho/km;  $C = 8 \times 10^{-6}$  F/km

| L (mH) | $Z_0$ (ohms)  | $\alpha$ (Np/km) | $\beta$ (rad/km) |
|--------|---------------|------------------|------------------|
| 3.5    | 21.05 – j2.36 | 0.23754          | 2.11613          |
| 6.0    | 27.45 – j1.81 | 0.18218          | 2.75917          |
| 8.5    | 32.63 – j1.52 | 0.15323          | 3.28049          |
| 11.0   | 37.11 – j1.34 | 0.13476          | 3.73022          |

#### OUTPUT SCREENSHOT:



## Test Case 3

Explore the impact of frequency and capacitance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line. Vary the frequency from 2 KHz to 10 KHz at intervals of 2 KHz with L, R, G and C constant, then vary the capacitance from 8  $\mu\text{F}$  to 16  $\mu\text{F}$  at intervals of 2  $\mu\text{F}$  with R, G and L constant.

### PROGRAM:

```
clear; clc;
R=10; L=0.009; f=2*10^3; G=4*(10^-6); C=16*(10^-6);
w=2*pi*f; Z=R+(%i*w*L); Y=G+(%i*w*C); Zo=sqrt(Z/Y);
C1=round(real(Zo)); D=round(imag(Zo));
printf("-Zo = %f + j(%f)ohms\n", C1, D); P=sqrt(Z*Y);
a=real(P); a1=round(a*10000)/10000;
printf("-Attenuation constant a = %f neper/km\n", a1); b=imag(P);
b1=round(b*10000)/10000;
printf("-Phase constant b = %f radians/km", b1);
```

### OUTPUT:

Variable f — Constants: L = 9 mH; R = 10  $\Omega$ ; G =  $4 \times 10^{-6}$  mho/km; C =  $16 \times 10^{-6}$  F/km

| f (kHz) | $Z_0$ (ohms)  | $\alpha$ (Np/km) | $\beta$ (rad/km) |
|---------|---------------|------------------|------------------|
| 2       | 23.74 – j1.05 | 0.21066          | 4.77325          |
| 4       | 23.72 – j0.52 | 0.21081          | 9.53953          |
| 6       | 23.72 – j0.35 | 0.21084          | 14.30736         |
| 8       | 23.72 – j0.26 | 0.21085          | 19.07557         |
| 10      | 23.72 – j0.21 | 0.21086          | 23.84394         |

Variable C — Constants: R = 10  $\Omega$ ; L = 9 mH; f = 2500 Hz; G =  $0.4 \times 10^{-6}$  mho/km

| C (μF) | Zo (ohms)     | α (Np/km) | β (rad/km) |
|--------|---------------|-----------|------------|
| 8      | 33.56 – j1.19 | 0.14898   | 4.21752    |
| 10     | 30.02 – j1.06 | 0.16657   | 4.71533    |
| 12     | 27.40 – j0.97 | 0.18247   | 5.16539    |
| 14     | 25.37 – j0.90 | 0.19708   | 5.57926    |
| 16     | 23.73 – j0.84 | 0.21069   | 5.96448    |

OUTPUT SCREENSHOT:

